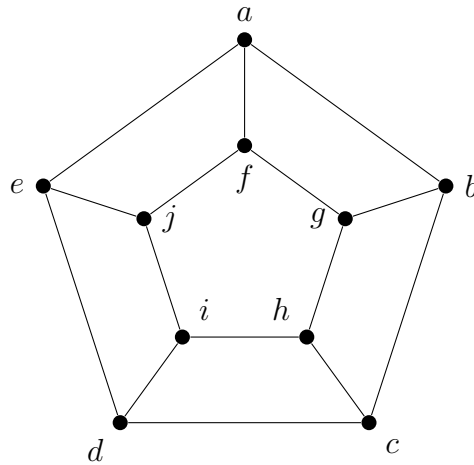
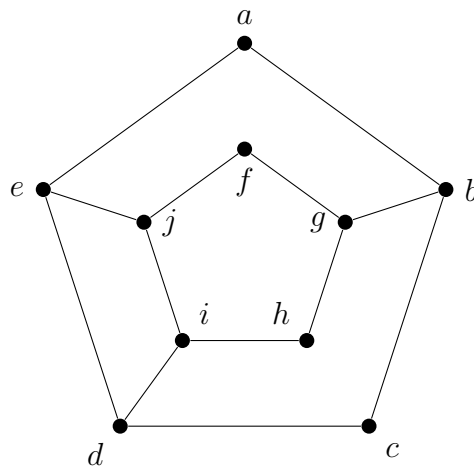


2.2

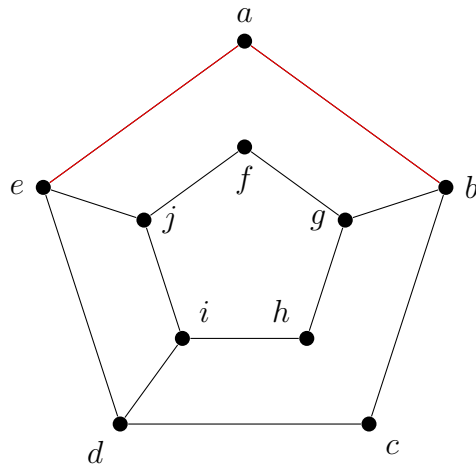
6. Show that there can be no Hamilton circuit in the following graph using both edges (a, f) and (c, h) .



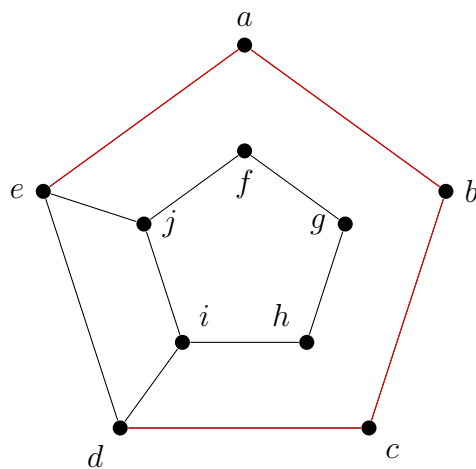
- **Delete** (a, f) and (c, h)



- **Force** (a, b) and (a, e) ($\deg(a) = 2$)

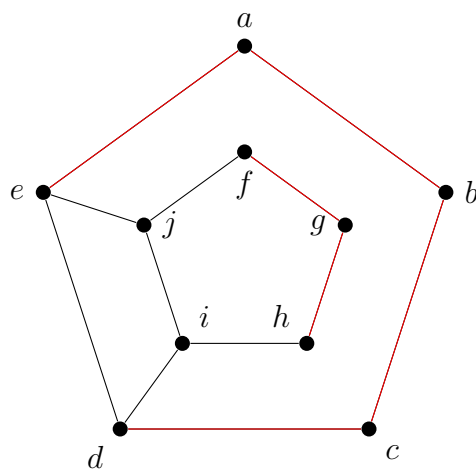


- **Force** (c, b) and (c, d) ($\deg(c) = 2$)

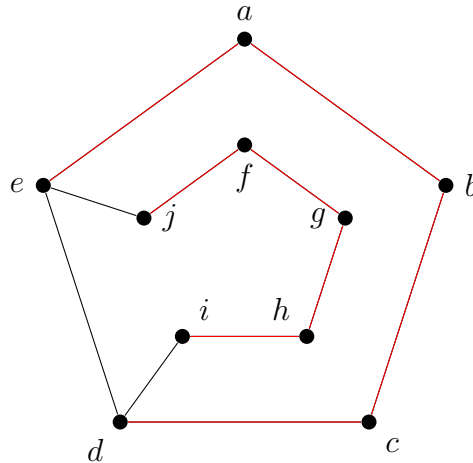


- Delete (b, g) ($\deg(b)$ is 2 after removal)

- **Force** (g, h) and (g, f) ($\deg(g) = 2$)

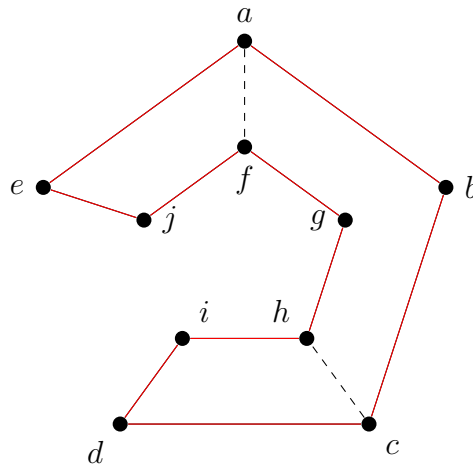


- **Force** to (f, j) and (h, i) ($\deg(f) = \deg(h) = 2$)



- Delete (j, i) (Creates the sub-circuit $f - g - h - i - j - f$)

Force (d, i) and (e, j) ($\deg(i) = \deg(j) = 2$)



- Delete (e, d) ($\deg(e)$ and $\deg(d)$ is 2 after removal)

$\exists$ a H.C: $a - b - c - d - i - h - g - f - j - e - a$.

$\Rightarrow$

Cannot use (a, f) and (c, h) because it would make $\deg(a), \deg(f), \deg(c)$ and $\deg(h) = 3$, which is impossible for a H.C.

9. Recall that a graph is *bipartite* if the vertices can be divided into two sets; for convenience call them blue vertices and red vertices, such that every edge connects a blue and a red vertex.

(a) Show that if a connected bipartite graph has a Hamilton circuit, then the number of red and blue vertices must be equal. Further, show that if a bipartite graph has an odd number of vertices, then it has no Hamilton circuit.

Proof. Suppose G is a connected bipartite graph with a Hamilton circuit. Let R and B be the independent sets of red and blue vertices in G . Since G has a Hamilton circuit and every edge connects a red and blue vertex, every vertex in G is not only reached but the

H.C alternates between a red and blue vertex. So, it must be that $|R| = |B|$. Suppose G contained an odd # of vertices. Since G is bipartite, every circuit in G must be of even length. But, a graph of odd vertices would create a circuit of odd length, which is impossible for a bipartite graph. Hence, a Hamilton circuit can not exist in a graph with an odd # of vertices. $\square$

(b) Show that if a connected bipartite graph has a Hamilton path, then the numbers of reds and blues can differ by at most one.

Proof. Let G be a connected bipartite graph that has a Hamilton path P . We know that P must reach every vertex in G . By contradiction, suppose the # of red and blue vertices differ by at least 2. Let R be the set of red vertices and B be the set of blue vertices and let $r_1 \in R$ and $b_1 \in B$. WLOG let $|B| \geq (|R| + 2)$.

Case 1: Suppose we start at r_1 such that

$$P = r_1 - b_1 - r_2 - b_2 - \dots - r_n - b_n.$$

But, since $|B| > |R|$, b_n is not the last vertex in B . So, vertices

$$b_{n+1}, b_{n+2}, \dots, b_{n+k} \text{ where } k = |R|$$

will be unreachable, contradicting P being a Hamilton path.

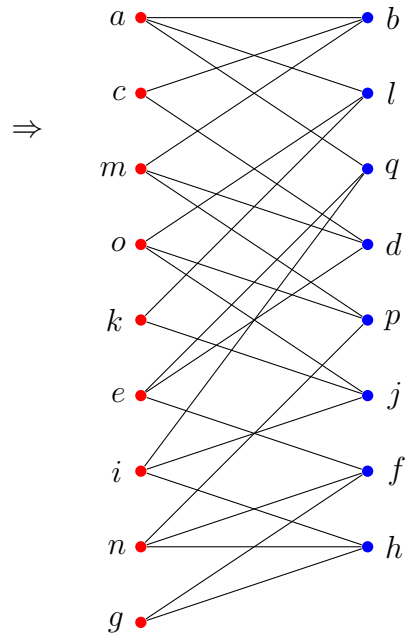
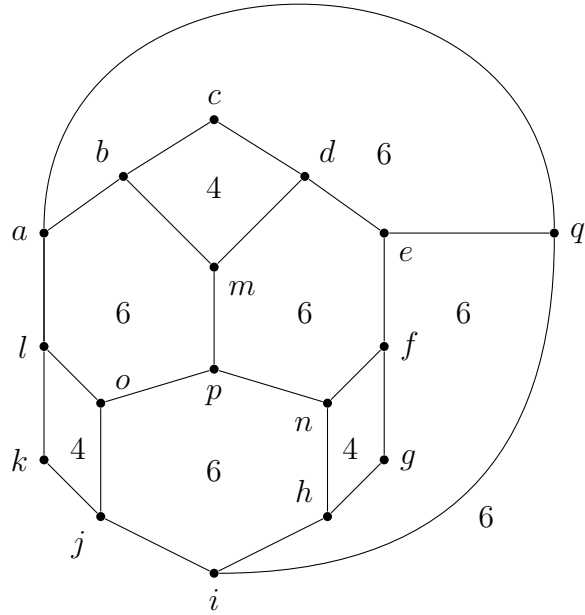
Case 2: Suppose we start at b_1 such that

$$P = b_1 - r_1 - b_2 - r_2 - \dots - b_n - r_n - b_{n+1}.$$

The same problem arises in which vertices after b_{n+1} will be unreachable, contradicting P being a Hamilton path. Hence, the difference of $|R|$ and $|B|$ must differ by at most 1 for G to have a Hamilton path. $\square$

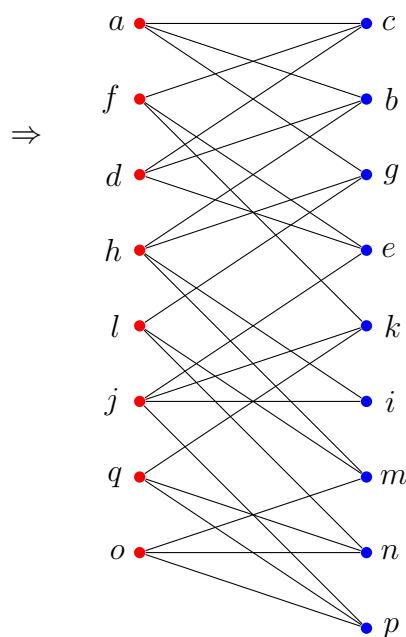
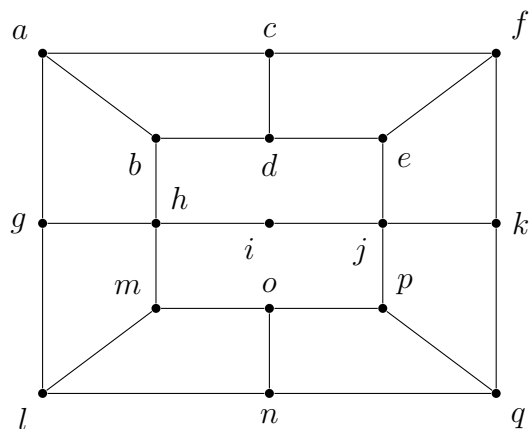
(c) Use part (a) to show that the following graphs have no Hamilton circuit:

(i)


$$\Rightarrow G \text{ is bipartite!}$$

But G has 17 vertices which is odd. So, no Hamilton circuit exists.

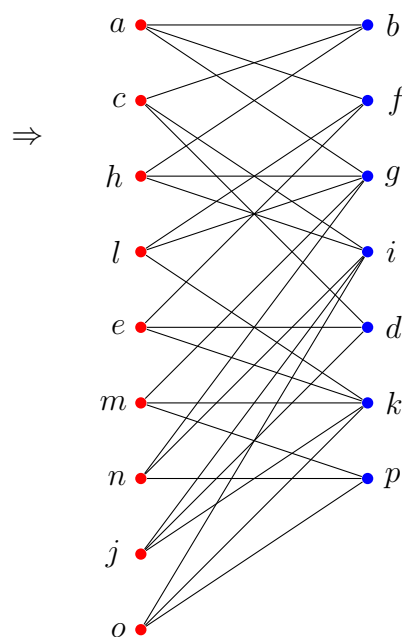
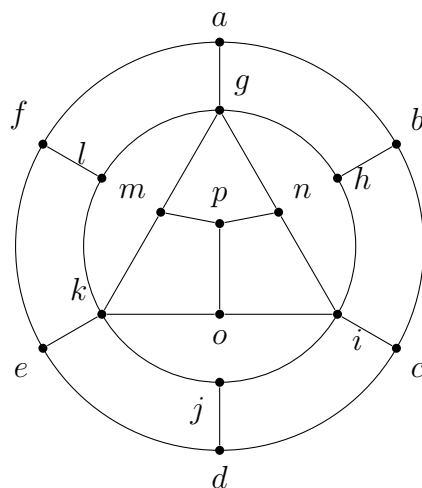
(ii)



$\Rightarrow G$ is bipartite!

But G has 17 vertices which is odd. So, no H. C exists.

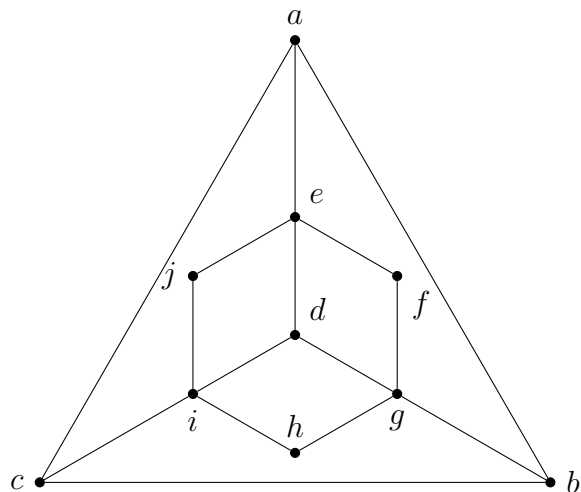
(iii)



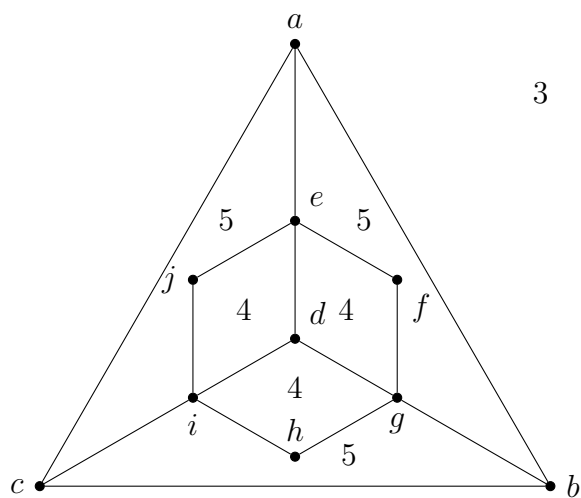
$\Rightarrow G$ is bipartite!

But $|R| = 9 \neq |B| = 7$. So, no H.C exists.

7(a). Prove that the following graph has no Hamilton circuit. (*Use Theorem 3)



- Is this graph G planar? yes



- Does this graph have a H.C?

$$\Rightarrow i = 3 \xrightarrow{1 \text{ region}} r_3 + r'_3 = 1$$

$$\Rightarrow i = 4 \xrightarrow{3 \text{ regions}} r_4 + r'_4 = 3$$

$$\Rightarrow i = 5 \xrightarrow{3 \text{ regions}} r_5 + r'_5 = 3$$

$$* \sum_i (i - 2)(r_i - r'_i) = 0$$

$$\Rightarrow (3-2)(r_3 - r'_3) + (4-2)(r_4 - r'_4) + (5-2)(r_5 - r'_5) = 0$$

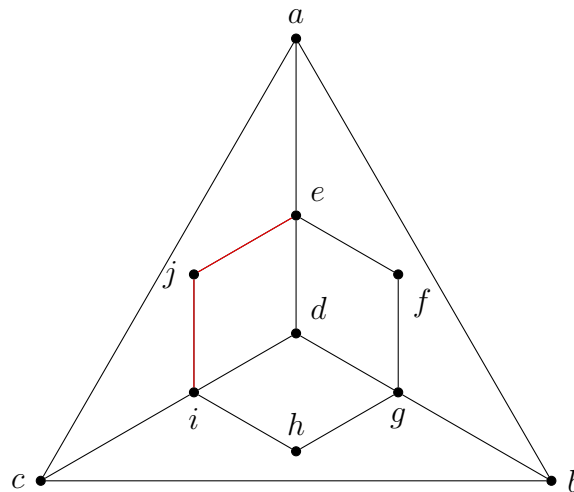
$$\Rightarrow (r_3 - r'_3) + 2(r_4 - r'_4) + 3(r_5 - r'_5) = 0$$

$$\Rightarrow \pm\{1\} \pm 2\{1,3\} \pm 3\{1,3\} = 0$$

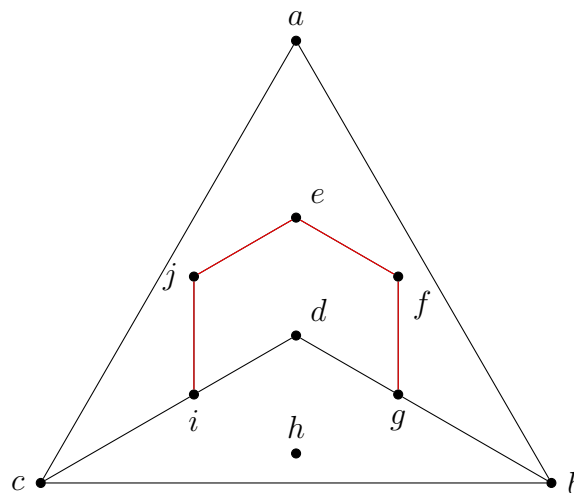
$$\Rightarrow \pm\{1\} \pm \{2,6\} \pm \{3,9\} = 0$$

$$\Rightarrow \text{inconclusive due to the possible solution } 1 + 2 - 3 = 0.$$

- **Force** (j, i) and (j, e) ($\deg(j) = 2$)



- **Force** (f, e) and (f, g) ($\deg(f) = 2$)



- Delete (e, d) and (e, a) ($\deg(e)$ is 2 after removal)

- Delete (h, g) and (h, i) because of the sub-circuit $e - f - g - h - i - j - e$.

Cannot be done because preventing a sub-circuit causes $\deg(h) = 0$, which is impossible in a H.C..

Exhausted all choices $\therefore$ no H.C exists
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